

1. Emergent Behavior

Neural networks exhibit sudden appearance of new abilities when scaled. These abilities are not explicitly programmed. This resembles phase transitions in physics, where small quantitative changes lead to qualitative differences. The mystery lies in predicting when and why these abilities appear.

2. Representation Learning

Neural networks transform raw inputs into latent representations. These representations capture abstract features and relationships. However, these features are distributed across neurons and not explicitly defined, making them hard to interpret.

3. Interpretability Problem

Deep models act as black boxes. While inputs and outputs are known, the internal reasoning process is unclear. Understanding decision pathways inside large models remains an open research challenge.

4. Training Dynamics

Gradient descent successfully optimizes highly non-convex loss landscapes. Despite the complexity, it consistently finds good solutions. Why it avoids poor minima and converges efficiently is not fully understood.

5. Loss Landscape Geometry

The loss surface of neural networks is extremely high-dimensional. Flat minima correspond to better generalization, while sharp minima do not. Understanding why optimization prefers flat regions remains an open question.

6. Generalization Paradox

Large models with more parameters than data points generalize well, contradicting classical learning theory. Implicit regularization during training may explain this, but a full theory is lacking.

7. Double Descent

Model error decreases, then increases, and then decreases again as model size grows. This breaks the classical bias-variance tradeoff and suggests new learning dynamics.

8. In-Context Learning

Transformers can learn tasks from examples in prompts without updating weights. This suggests internal simulation of learning processes, but the mechanism is still unclear.

9. Mechanistic Circuits

Specific circuits inside networks perform identifiable computations. However, mapping these circuits systematically is extremely difficult.

10. Adversarial Examples

Small, imperceptible changes to input can drastically change model output. This reveals sensitivity to features that humans do not perceive.

11. Grokking Phenomenon

Models initially memorize data and later suddenly generalize. This delayed understanding suggests internal restructuring over time.

12. Scaling Laws

Performance improves predictably with scale (data, parameters, compute). The mathematical reason for these power-law relationships is not fully understood.

13. Neural Collapse

Late in training, representations become highly structured and symmetric across classes. This geometric organization emerges naturally but lacks a complete explanation.

14. Sparse vs Dense Knowledge

Knowledge in neural networks can be both distributed and localized. Understanding how information is stored remains unclear.

15. Alignment Problem

Models optimize for mathematical objectives, but desired human-aligned behavior is complex. Ensuring models act as intended remains a major challenge.